Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/04/2015
PI Name:	Rojas-Pierce, Marcela Ph.D.		
Project Title:	The Role of Vacuole Membrane Fusion in Plant Gravity Perception		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	27695-7612	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/02/2013	**End Date:**	07/01/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 7/1/2017 per NSSC information (Ed., 4/5/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AM49G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The long-term goal of this research is to identify and characterize mechanisms of plant vacuole biogenesis that are critical for gravity perception. A role of the vacuole in controlling gravity perception is well accepted, but the molecular mechanisms by which the vacuole interacts with amyloplasts or regulates their sedimentation during gravitropism are yet to be identified. We recently uncovered a novel vacuole phenotype in agravitropic mutant alleles of VTI11 in Arabidopsis. We predict that this vacuole phenotype of vti11 mutants has negative effects on the organization of actin cables or the cortical endoplasmic reticulum, and therefore prevents amyloplast sedimentation. In addition, we have uncovered a novel regulation of vacuolar membrane fusion by phosphoinositides. Based on our results, we developed the hypothesis that mechanisms of vacuole remodeling that are mediated by phosphoinositide signaling are critical for the perception or transduction of the gravitropic signal in plants. A unique assay for vacuole fusion that we developed		

	will be used to determine the role of specific cellular components in membrane fusion and gravity perception. The specific aims of this research project are to: 1) Determine subcellular phenotypes that impair gravity perception in itt3/vti11. 2) Identify the role of phosphoinositides in gravitropism. 3) Determine the role of the Adaptor Protein (AP-3) complex on vacuole biogenesis and gravitropism. Overall, this research will improve our understanding of vacuole biogenesis, the interactions between the vacuole and other cellular components involved in gravity perception, and the role of phosphoinosidides in gravitropism.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will contribute to the understanding of the molecular mechanisms that control the response of plants to gravity. In addition, the function of the vacuole in establishing the morphology and dynamics of other organelles will be characterized. These results will enhance the fundamental understanding of how plant cells organize into a defined set of independent organelles.
Task Progress:	Personnel Jiameng Zheng is a fifth year graduate student. Jiameng was involved in the identification and characterization of vps41 loss of function alleles, generation of many of the clones for protein expression and localization in plants, and for protein-protein interaction analysis. She is the first author in a small publication where we showed that the itt3 vacuoles are independent organelles. In addition to conducting research, Jiameng presented a poster and gave a talk at the Mid West Cell Biology meeting in May 2014. One postdoctoral scholar, Dr. Ashley Alvarez, has been involved in gravity perception studies in Arabidopsis. She has used the vertical microscope to quantify amyloplast sedimentation in itt3 and zig-1 mutants under different conditions. She has also analyzed organelle phenotypes in these mutants that result from abnormal vacuoles, and generated a suite of markers for visualizing cellular organelles in the shoot endodermis of Arabidopsis. Two undergraduate students were involved in this research during the reporting year. Kimberly Baker assisted with technical support and gained some experience with molecular biology. Kimberly graduated in May 2014 with a B.Sc. in Plant Biology. Jennifer Kalinowski joined the lab in Fall 2014 and is a student in the Plant Biology program as well. She is continuing to get research experience in the laboratory. Research progress The specific aims of this research project are to: 1) Determine subcellular phenotypes that impair gravity perception in itt3/vti11. 2) Identify the role of phosphoinositides in gravitropism. 3) Determine the role of the Adaptor Protein (AP-3) complex on vacuole biogenesis and gravitropism. We have made good progress on this project during the second year of the award: a) We published an article that demonstrates that the vacuoles of vti11 mutants are independent organelles. Jiameng Zheng, a graduate student supported on this grant since Fall 2013, is the first author in this publication. b) We have introgressed a series of subcellular organelles into vti11 alleles to determine the effect of the fragmented vacuole phenotype on the organization and function of other cellular compartments or structures. We have determined that the morphology of the vacuole has dramatic consequences for cellular organization, but this effect is restricted to a subset of cellular compartments. c) We have used our unique vertical stage microscope for live-cell imaging of amyloplasts in wild type and in vit11 mutant alleles. We have determined that, similar to the inflorescence stem, sedimentation of amyloplasts from vti11 mutant hypocotyls is severely restricted. Treatment with Wortmannin resulted in a significant increase in amyloplast sedimentation in these cells as a result of vacuole fusion. Therefore, we conclude that the fragmented vacuole phenotype of vti11 mutants is the cause of their agravitropic phenotype, and that mechanisms of vacuole dynamics may regulate gravi-perception. d) The endodermis was previously identified as the cell layer responsible for gravi-perception in shoots. As part of this work, we determined that the entire endodermis along Arabidopsis hypocotyls does not display amyloplast sedimentation and therefore could not be gravi-perceptive. We have identified specific regions of the hypocotyl endodermis that are responsible for the perception of gravity signals. Our work is now focused on these cells of the hypocotyl.
Bibliography Type:	Description: (Last Updated: 11/17/2022)
Abstracts for Journals and Proceedings	Zheng J, Han SW, Rodriguez-Welsh MF, Rojas-Pierce M. "Phosphatidylinositol 3-Phosphate is a negative regulator of vacuolar fusion in plants." Plant Systems and Synthetic Biology: Solutions to Global Food Security Symposium, Raleigh, NC, October 18-19, 2014. Plant Systems and Synthetic Biology: Solutions to Global Food Security Symposium, Raleigh, NC, October 18-19, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Alvarez A, Rojas-Pierce M. "Vacuole membrane fusion and dynamics may play a role in Arabidopsis gravity perception." Poster at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. Abstract book. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Brillada C, Zheng J, Rojas-Pierce M. "The role of HOPS tethering complex in plant vacuole fusion." Poster at the 25th International Conference on Arabidopsis Research 2014, Vancouver, Canada, July 28-August 1, 2014. Abstract book. 25th International Conference on Arabidopsis Research 2014, Vancouver, Canada, July 28-August 1, 2014. , Jul-2014

Abstracts for Journals and Proceedings	Zheng J, Han SW, Rodriguez-Welsh MF, Rojas-Pierce M. "Phosphatidylinositol 3-Phosphate is a negative regulator of vacuolar fusion in plants." Poster at the Midwest Plant Cell Dynamics Meeting, Madison, WI, June 4-6, 2014. Midwest Plant Cell Dynamics Meeting, Madison, WI, June 4-6, 2014. , Jun-2014
Articles in Peer-reviewed Journals	Zheng J, Han SW, Munnik T, Rojas-Pierce M. "Multiple vacuoles in impaired tonoplast trafficking3 mutants are independent organelles." Plant Signaling and Behavior. 2014 Dec;9(10):e972113. http://dx.doi.org/10.4161/psb.29783 ; PubMed PMID: 25482812 , Dec-2014